Nervous System Answer Key

Multiple Choice Answers:

1. B) 12
2. C) Auditory cortex in the temporal lobe
3. B) Transmits signals between different brain regions
4. C) Thoracic and lumbar
5. B) Send motor signals from the brain to muscles and glands
6. C) Somatic is voluntary; autonomic is involuntary
7. C) Both produce myelin
8. B) Peripheral nervous system
9. C) Increased impulse speed
10. B) Nodes of Ranvier
11. B) adjacent
12. C) Presynaptic terminal and synaptic vesicles
13. B) Tyrosine
14. B) To act as a natural painkiller by inhibiting pain signals
15. A) Central nervous system
16. C) It is maintained by tight junctions between endothelial cells of brain capillaries.
17. D) Dopamine - AMPA receptors in motor control pathways.
18. A) It increases the resting membrane potential, inactivating voltage-gated sodium channels.

Short Answer Questions:

Roles and parts of the hindbrain (3 pts):

1. Medulla oblongata - Controls autonomic functions like heart rate, blood pressure, and breathing.

Pons - Relays information between the cerebrum and cerebellum; helps regulate sleep and respiration.

Cerebellum - Coordinates voluntary movements, balance, and posture.

If all 3 are only listed without defining, 1.5 pt
Any part named and defined is 1 pt

2. Three layers of the meninges (deep to superficial): (3 pts)

Pia mater

Arachnoid mater

Dura mater

Wrong order 2 pts
2 listed with correct order 1.5 pts
2 listed incorrect order 1 pt
1 listed 0.5 pts

3. Divergent integration (2 pts):

A single presynaptic neuron sends signals to multiple postsynaptic neurons, amplifying or spreading the signal to multiple pathways or responses.

1.0 pt - States that one presynaptic neuron communicates with multiple postsynaptic neurons.

1.0 pt - Mentions that the effect is to spread or amplify the signal.

4. Four (possible) cranial nerves and their physiology (4 pts):

CN I (Olfactory): Responsible for smell (sensory).

CN II (Optic): Transmits visual information from the retina (sensory).

CN VII (Facial): Controls facial expressions, taste from anterior 2/3 of tongue, and some glandular secretions (mixed).

CN X (Vagus): Regulates parasympathetic control of heart, lungs, and digestive tract (mixed).

Per nerve:
0.5 pt - Correct nerve name and Roman numeral.

0.25 pt - Correct general function.

0.25 pt - Correct physiological description (lenient).

5. Neurotransmitter Categorization:

Epinephrine - Catecholamine

Histamine - Indolamine

Norepinephrine - Catecholamine

Dopamine - Catecholamine

Serotonin - Indolamine

6. Matching (Terms to Definitions):

D. Parkinson's Disease - Progressive neurological disorder causing tremors, rigidity, and slow movement due to loss of dopamine-producing brain cells.

A. Epilepsy - Recurrent, uncontrolled seizures from abnormal brain activity.

C. Multiple Sclerosis (MS) - Immune system attacks myelin sheath, disrupting nerve communication.

E. Cerebral Palsy - Group of disorders affecting movement/posture due to brain damage before/after birth.

B. Alzheimer's Disease - Progressive memory loss and cognitive decline due to brain cell degeneration.

F. Shingles (Herpes Zoster) - Viral infection with painful rash/blisters on one side of the body.

G. Stroke - Blood flow to the brain is interrupted, causing brain cell damage.

H. ALS - Progressive motor neuron disease causing muscle weakness and paralysis.

7. Neuron Labelling

A. Dendrite
B. Cell body
C. Axon
D. Myelin Sheath
E. Node of Ranvier
F. Synapse

8. Chemical Structures Question:

- Glycine, GABA, aspartic acid, glutamic acid (amino acid neurotransmitters)

- Quarternary ammonium group with a positively charged nitrogen atom, which allows for binding to acetylcholine receptors, specifically muscarinic and nicotinic
- The negatively charged chlorine atom improves the solubility and bioavailability of acetylcholine, and makes it easy to create salts for pharmaceutical use.

9. Brain Labelling:

A. Frontal lobe
B. Parietal lobe
C. Occipital lobe
D. Temporal lobe
E. Cerebellum
F. Brainstem (medulla)
G. Corpus callosum
H. Thalamus
I. Hypothalamus
J. pituitary gland
K. Pineal gland

10. Without inhibitory neurotransmitters, excitatory neurotransmitters will overstimulate nerve cells in the brain.

Ex: say that the neurotransmitter epinephrine begins to stimulate the brain due to a drug. Because of an excessive amount of epinephrine present, serotonin should come in and begin its inhibitory process. If it does not, epinephrine begins to over excite the nerve cells which can result in negative effects.

Mentions excitatory neurotransmitters overstimulating neurons - 1.5 pts
Any applicable neurotransmitter can be used as an example, but the mechanism of action of the excitatory neurotransmitter must be detailed at least at a basic level for 1.5 pts.

11. Case Study:

Likely diagnosis: Multiple sclerosis

Pathophysiology: Autoimmune attack on CNS myelin (by T cells and macrophages) damages oligodendrocytes. Myelin normally increases membrane resistance and decreases capacitance, enabling rapid saltatory conduction. Demyelination slows or blocks action potentials.

Heat exacerbation: Increased temperature further impairs conduction in partially demyelinated axons (Uhthoff's phenomenon).

Partial Credit Breakdown:

1 pt - Correct diagnosis (MS).

1.5 pts - Accurate explanation of myelin's role in conduction and how demyelination disrupts it.

1 pt - Explanation of why heat worsens symptoms (Uhthoff's phenomenon).

0.5 pt - Mentions autoimmunity or oligodendrocyte damage but not fully explained.

Sense Organs Answer Key

Multiple Choice

1. D
2. C
3. B
4. A
5. C
6. B
7. B
8. B
9. A
10. D
11. A
12. C
13. C
14. C
15. C
16. B
17. A
18. C
19. B
20. B
21. B
22. C
23. C

Free Response:

1.

Full Credit Response will include:

Photon, retina, opsin activation, cascade, hyperpolarization of photoreceptor, bipolar cell depolarization, ganglion cell action potential, optic nerve.

Partial credit:

- 2 pts for explaining photopigment change but missing downstream steps.
- 2 pts for explaining hyperpolarization but missing molecular pathway.
- 1 pt for mentioning optic nerve transmission without explaining preceding events.

2.

Full Credit Response will include:

Pitch is determined by place theory, tonotopic organization maintained through cochlear nerve, brainstem nuclei, and primary auditory cortex frequency map

Loudness is determined by the amplitude of basilar membrane displacement, number of hair cells stimulated, firing rate and recruitment of neurons, and intensity coding in auditory cortex
Partial Credit Response:

- 2 pts for distinguishing pitch vs. loudness in the cochlea only.
- 2 pts for explaining central processing differences but not cochlear mechanisms.
- 1 pt for noting tonotopy but not connecting it to pitch.

3.
Full Credit Response will include:
Produces left homonymous hemianopia, loss of visual field in left half of both eyes (nasal retina of right eye, temporal retina of left eye), contralateral effect due to partial decussation at optic chiasm.
Partial credit:

- 2 pts for identifying correct visual field loss but not explaining anatomy.
- 1 pt for identifying contralateral field loss but wrong side.

4.
Full Credit Response will include:
Odorant binds to GPCR on olfactory receptor neurons, activation of Golf, rising of cAMP, opening of cation channels, depolarization, signal sent to olfactory bulb, mitral cells, olfactory tract, direct projection to primary olfactory cortex (piriform) and amygdala, connections to hippocampus and limbic system mediate emotion/memory.
Partial credit:

- 2 pts for receptor-level events without central connections.
- 2 pts for central pathways but no receptor events.
- 1 pt for mentioning limbic link without pathway detail.

5.
Full Credit Response will include:
Degeneration of macula (central retina), drusen deposits, atrophy of retinal pigment epithelium, or abnormal choroidal neovascularization (wet type). Risk factors include age, smoking, family history, cardiovascular disease, obesity.
Treatment: Anti-VEGF injections (wet AMD), photodynamic therapy, lifestyle changes, antioxidant supplementation (AREDS formula).
Partial credit:

- 2 pts for pathophysiology only.
- 2 pts for listing correct risk factors.
- 1 - 2 pts for mentioning any valid treatment method.

6.

Full credit:
Clear drawing with scala vestibuli, scala media, scala tympani, organ of Corti, basilar membrane, tectorial membrane. Frequency mapping: high frequencies near base, low near apex.
Partial credit:

- 2 pts for correct structures without frequency mapping.
- 1 - 2 pts for frequency mapping on incorrect/missing structures.

7.
Full Credit Response will include:
Sense of body position and movement, using muscle spindles, Golgi tendon organs, and joint receptors.
Partial credit:

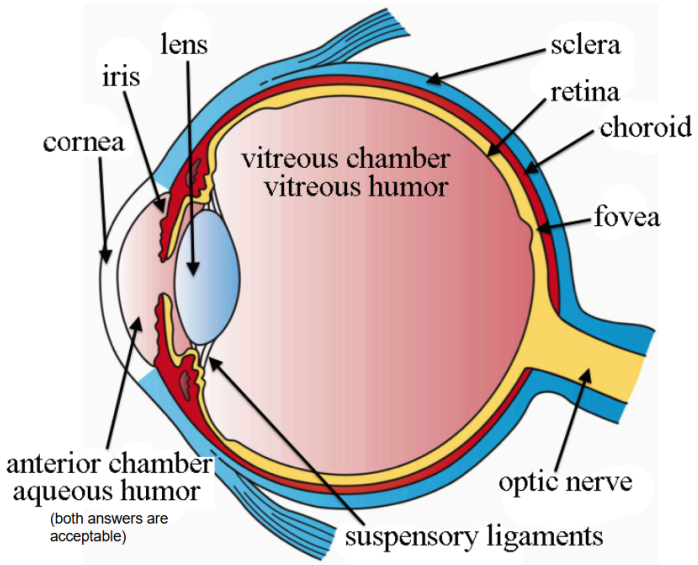
- 1 pt for defining as “sense of body position” without receptors.

8.
Full Credit Response will include:
General senses: Touch, pressure, pain, temperature, proprioception-widely distributed.
Special senses: Vision, hearing, taste, smell, equilibrium-localized organs.
Partial credit:

- 1 pt for listing correct examples without explaining distribution/localization.

Matching:
H – Inflammation of the middle ear
D – Difficulty or inability to see in low light
K – Loss of the ability to smell
B – Farsightedness
I – Hearing loss from inner ear/auditory nerve damage
J – Hearing loss from outer/middle ear issue
E – Irregular cornea/lens curvature
F – Pink eye
C – Age-related near vision loss
L – Distorted sense of smell
A – Nearsightedness
G – Inability to distinguish colors
M – Distorted taste

Diagram:


lens
iris
cornea
sclera
retina
choroid
vitreous chamber
vitreous humor
fovea
anterior chamber
aqueous humor
(both answers are acceptable)
optic nerve
suspensory ligaments

Endocrine System Answer Key

Multiple Choice

1. B
2. C
3. B
4. B
5. A
6. C
7. C
8. C
9. B
10. D
11. B
12. B
13. B
14. B
15. A
16. D
17. C
18. A
19. B
20. B
21. C
22. C
23. C

Free Response:

1.

Full credit response will include:

Blood bound to carrier proteins, diffuses through cell membrane, binds intracellular receptor, hormone-receptor complex, nucleus, binds DNA, gene transcription.

Partial Credit: Deduct 1–2 pts for missing intermediate steps but showing correct sequence logic.

2.

Full Credit Response:

Feature	Primary	Secondary
Cause	Adrenal cortex failure	Low ACTH secretion
Cortisol	Decreases	Decreases

ACTH	Increases	Decreases
Aldosterone	Decreases	Normal
Hyperpigmentation	Present	Absent
Electrolytes	Hyponatremia, hyperkalemia	Often normal K+

Partial credit: Give 1 pt for each correctly stated row in the table.

3.
Full Credit response wil include:
Hypothalamus links CNS signals to endocrine output via neurosecretory cells.
Example 1: CRH from hypothalamus, ACTH from pituitary, cortisol from adrenal cortex (stress response).
Example 2: TRH, TSH, thyroid hormones.
Receives neural input from limbic system, brainstem, and senses (e.g., light → SCN → circadian rhythms)
Partial credit: Half points if pathway listed but hormone order or target organ wrong.

4.
Full credit will include:
Hypothalamus releases TRH, pituitary releases TSH, thyroid releases T_3/T_4.
High T_3/T_4 feeds back to inhibit TRH & TSH secretion.
Primary hyperthyroidism: Thyroid overproduces T_3/T_4 independent of TSH, decreasing TSH, decreasing TRH, increasing T_3/T_4.
Partial credit: Deduct 1–2 pts if TRH feedback omitted but TSH prediction correct.

5.
Full credit answer will include:
High LH:FSH ratio (often > 2:1).
Hyperandrogenism from theca cells, hirsutism, acne
Insulin resistance, rising insulin, rising androgen production.
Anovulation, menstrual irregularities, infertility.
Possible rising levels of estrogen from peripheral conversion, endometrial hyperplasia risk.
Partial credit: 1 pt for each correct hormone change or clinical consequence.

6.
Full credit answer will include
Hypothalamusm, CRH release, anterior pituitary, ACTH, adrenal cortex, cortisol, Sympathetic activation, adrenal medulla, epinephrine/norepinephrine, Cortisol effects rising, glucose, rising

BP, immune modulation, Catecholamine effects: rising HR, rising BP, rising glucose, high alertness.
Partial credit:
Give half points for naming hormones in order without their sources.
Deduct 1–2 pts if only one arm of stress response included.

Diagram:

